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Studierichting: Informatica
Oefeningenreeks 5F nr. 4

$$9y^2 = x(x-3)^2$$

Symmetrie: $y^2 \implies$ symmetrisch t.o.v. de x -as

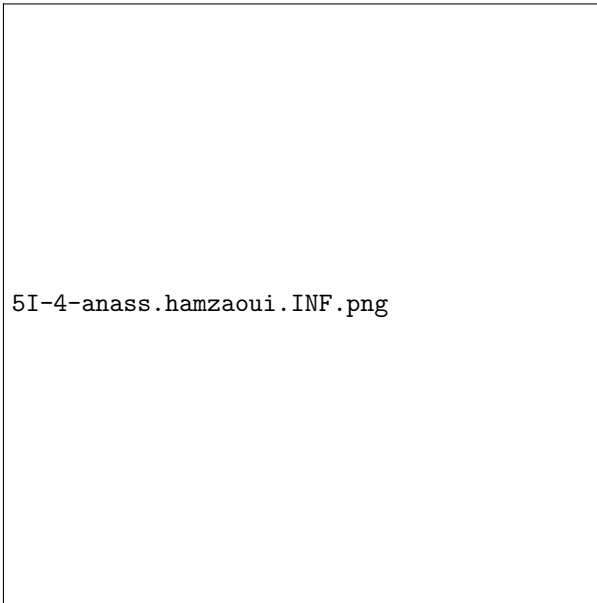
Domein: $x(x-3)^2 \geq 0 \implies x \geq 0$

Snijpunten met x -as: $y = 0 \implies x(x-3)^2 = 0 \implies x = 0$ en $x = 3$

Bovenste helft: $y = \frac{1}{3}(3-x)\sqrt{x} = \sqrt{x} - \frac{1}{3}x^{3/2}$

Maximum: $y' = 0 \implies x = 1 \implies \left(1, \frac{2}{3}\right)$, minimum: $\left(1, -\frac{2}{3}\right)$

Grafiek:



$$\begin{aligned}
L &= 2 \int_0^3 \sqrt{1 + (y')^2} \, dx \\
y &= x^{1/2} - \frac{1}{3}x^{3/2} \implies y' = \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{1/2} = \frac{1-x}{2\sqrt{x}} \\
(y')^2 &= \frac{(1-x)^2}{4x} = \frac{1-2x+x^2}{4x} \\
1 + (y')^2 &= \frac{4x+1-2x+x^2}{4x} = \frac{(x+1)^2}{4x} \\
\sqrt{1 + (y')^2} &= \frac{x+1}{2\sqrt{x}} = \frac{1}{2}x^{-1/2} + \frac{1}{2}x^{1/2} \\
L &= 2 \int_0^3 \left(\frac{1}{2}x^{-1/2} + \frac{1}{2}x^{1/2} \right) dx \\
&= 2 \left[x^{1/2} + \frac{1}{3}x^{3/2} \right]_0^3 \\
&= 2 \left(\sqrt{3} + \frac{1}{3} \cdot 3\sqrt{3} \right) \\
&= 2 \cdot 2\sqrt{3} = 4\sqrt{3}
\end{aligned}$$

References